Ambient Systems

IoT Appendix

The Ambient Systems IoT offering integrates robust hardware and software solutions for real-time environmental data acquisition, designed for use in dynamic operational environments. Here’s a summary of each component and their unified purpose:

1. Edge Acquisition System
2. IoT Communication Infrastructure
3. and Analytics Integration

Edge Acquisition System

Purpose: Collects and transmits environmental data (e.g., temperature, humidity, CO2 levels) from various locations to provide real-time insights.

Components:

- **Controller**: Central processing unit that aggregates sensor data.
- **Sensors**: Multiple sensors including CO2, temperature, humidity, and light sensors capture ambient conditions.
- **Wireless Connectivity**: 4G router for remote access and real-time data sharing.

Applications: Ideal for environments that require continuous monitoring of ambient conditions, such as vertical farming, commercial buildings, and logistics.

IoT Communication Infrastructure

Purpose: Provides the backbone for data transmission between edge devices and , ensuring seamless communication and secure data handling.

Components:

- **WiGig and Wi-Fi Modules**: High-speed, low-latency wireless connections between sensors, local storage, and the cloud.
- **Industrial PCs**: Edge devices equipped with caching capabilities and high storage capacity for local data retention.
- **MMWave and Cellular Connectivity**: Allows flexibility in urban and rural settings, adapting to different network availabilities.

Applications: Supports real-time data streaming in environments such as transportation, industrial facilities, and agricultural sectors.

and Analytics Integration

Purpose: Centralizes data storage and analytics, enabling robust data analysis and insights generation.

Components:

- **Cloud Server**: Aggregates data from edge devices for long-term storage and analysis.
- **Customer Platform Server**: Delivers processed insights and recommendations back to clients through an intuitive user interface.

Applications: Useful for large-scale data processing and actionable insights, offering predictive analytics for industries like commercial real estate, manufacturing, and resource management.

Who it’s for?

Use Case	Industry	Key Features	Benefits
Vertical Farming	Agriculture	IoT sensors for climate, humidity, and soil data; real-time analytics; edge processing	Optimize crop yields, reduce energy and water usage, and improve resource allocation
Fleet Management	Transportation, Logistics	GPS tracking, telematics data streaming, real-time vehicle health monitoring, mmWave communication units	Improve route efficiency, reduce fuel costs, and enhance vehicle maintenance practices
Smart Infrastructure	Urban Development, Real Estate	Environmental sensors, data aggregation for infrastructure health, predictive maintenance	Enhance infrastructure longevity, improve safety, and reduce repair costs through early problem detection
Manufacturing	Industrial Automation	Data capture from machinery, for low-latency processing, cloud analytics for long-term insights	Minimize downtime, improve production efficiency, and reduce waste through predictive maintenance
Environmental Monitoring	Environmental Services	Remote sensors for air quality, temperature, and humidity; cloud data storage for historical analysis	Provide accurate environmental data, monitor climate changes, and assist in regulatory compliance
Energy Management	Utilities	Energy usage tracking, predictive analytics, real-time monitoring of power systems	Reduce energy consumption, optimize power distribution, and improve sustainability initiatives
Real Estate Analytics	Commercial Real Estate	Occupancy tracking, financial performance metrics, sensor-based energy monitoring	Improve tenant retention, optimize space usage, and reduce operational

Use Case	Image Ideas
Vertical Farming	- Rows of indoor plants with IoT sensors monitoring temperature, humidity, and light levels in a controlled environment. - Dashboard displaying real-time data on crop health, water usage, and environmental conditions.
Fleet Management	- Vehicle fleet with mounted sensors and GPS, illustrating real-time data streaming to an operations center. - Interface showing route optimization and alerts for maintenance based on telematics data.
Smart Infrastructure	- Bridge or building with IoT sensors detecting structural integrity (e.g., vibration or temperature sensors). - Data visualization indicating maintenance needs, showing potential issues before they become critical.
Manufacturing	- Production line machinery with embedded sensors capturing performance metrics and maintenance data. - Real-time dashboard showing machinery health, efficiency, and alerts for preventive maintenance.
Environmental Monitoring	- Remote outdoor environment with air quality, temperature, and humidity sensors in various locations (e.g., urban or forested areas). - Map overlay visualizing air quality and climate data points over time.
Energy Management	- Utility facility or power grid with sensors monitoring energy consumption and flow, with an interface showing usage trends and efficiency metrics. - Dashboard showing predictive analytics on peak energy demands and potential savings.
Real Estate Analytics	- Office or retail space with occupancy sensors, climate control devices, and energy consumption metrics displayed on a building management dashboard. - Interface showing tenant satisfaction metrics tied to energy and comfort levels.
Smart City	- City streets with smart traffic lights, energy-efficient street lighting, and public safety cameras, all connected

Product Images

Hardware Images

1. **Industrial PC**: A rugged, compact computer with multiple ports for data input/output.
2. **Wi-Fi/Cellular Connectivity Modules**: Small, mountable devices enabling remote data streaming.
3. **IoT Sensors**: Modular components like cameras, GPS units, and environmental sensors, customizable for different needs.
4. **Cloud Server Architecture**: Visual representation of data flow from devices to and processing layers.

Component	Description	Potential Use Cases	Sample Image Description
Industrial PC	A durable, on-site computer (typically running Ubuntu) that handles local data processing, caching, and integration with other devices.	Vertical farming, Manufacturing, Fleet Management	Rugged, compact PC with multiple USB and Ethernet ports, designed to withstand industrial conditions.
Wi-Fi/Cellular Connectivity	Enables real-time data transmission to the cloud, supports WiGig/Wi-Fi, and cellular for remote locations.	Environmental Monitoring, Fleet Management	Small, mountable device with Wi-Fi/5G antennas for long-range data communication.
IoT Data Streamer	Streams real-time data from sensors and industrial machines, supporting integration with diverse IoT sensor networks.	Smart Infrastructure, Real Estate	Module connecting various sensors, data cables, and communication ports, facilitating smooth data transfer to processing units or .
IoT Sensors	Includes cameras, accelerometers, GPS, and environmental sensors for capturing a wide range of metrics (e.g., temperature, humidity, location).	Agriculture, Fleet Management, Environmental Safety	Modular, mountable sensors that are customizable based on data requirements, such as temperature and humidity sensors or cameras.
	server that collects, stores, and analyzes data for long-term insights, benchmarking, and historical analysis.	Real Estate, Vertical Farming	Data flow diagram showing connectivity from on-site hardware to and servers, supporting remote access and data analysis features.
File Server (Cache)	Local caching of data to ensure redundancy and reliable performance, especially when connectivity is intermittent or data volumes are high.	Smart Infrastructure, Manufacturing	A small, rugged device located near primary data sources (e.g., industrial machines), with high-speed data storage capabilities.
mmWave Communication Units	High-speed data transfer modules (e.g., 60 GHz units) for rapid data offloading from mobile assets like vehicles, ensuring data reaches cloud/storage in real-time.	Fleet Management, Environmental Monitoring	Compact devices with external antennas and high bandwidth capabilities, mounted on vehicles or in industrial environments.

Success Story / Example (Singapore Vertical Farm)

Example with yield and patent

Not servicing IoT solutions in Singapore anymore (no boots on the ground).

But we can provide this at *your* location.

Combine it with the Ambient FinTech platform

Ambient Systems IoT solution

The Ambient Systems IoT solution offers a comprehensive, end-to-end platform that combines environmental data acquisition, secure data communication, and centralized cloud-based analytics. This system supports industries requiring real-time environmental monitoring and automated insights to optimize operations and reduce costs. It is customizable to various environments and scalable for large deployments, making it adaptable for diverse sectors including agriculture, transportation, and commercial real estate.

Future Developments: IoT platform

1. Enhanced Data Processing on the edge
2. AI-Driven Insights on the edge
3. Modular Hardware Options
4. Integration with other hardware